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Ergonomic Evaluation of Different Surgeon Positions for Total Knee Arthroplasty Surgery

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Abstract: Ergonomics and risk factors for work-related musculoskeletal disorders have been studied extensively in various industry fields. However, only a few decades ago, these issues became a concern in the healthcare sector. Total knee arthroplasty (TKA) is one of the most common procedures performed by orthopaedic surgeons, and it would be desirable to perform it with an ergonomically safer technique. This study evaluated the ergonomic risk of different surgeon positions when performing contralateral TKA using the dominant hand. After the authors defined the four possible surgeon positions according to the most common positions used by surgeons in our environment (position A, on the opposite side of the knee to be operated on; position B, on the same side as the knee to be operated on; position C, with the patient's legs separated and the surgeon standing between them; and position D, facing the knee to be operated on, at the patient's feet), we performed an ergonomic analysis using the Rapid Entire Body Assessment (REBA) method. The overall REBA scores (lower score values indicate better ergonomics than higher) were between 7 and 6.5 for position A, between 6.17 and 5.5 for position B, between 5.92 and 5.5 for position C, and between 3.75 and 3.42 for position D. The test–retest and inter-rater reliability values ranged from substantial agreement to almost perfect agreement. Based on the results, we can conclude that the most ergonomic position for a right-handed surgeon to perform a left TKA is facing the left knee, at the patient's feet (position D).

Keywords: musculoskeletal disorders; ergonomic assessment; orthopaedic surgery; total knee arthroplasty

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1. Introduction

Among the occupational risks faced by orthopaedic surgeons (e.g., radiation exposure, infectious diseases, and chemical agents), one of the most common and least studied is ergonomic risk [1–4]. Orthopaedic surgery is a speciality with a particular risk of musculoskeletal disorders [5] because the orthopaedic surgeon's usual activities require physical strength, forced postures, repetitive movements, and prolonged standing [3,6]. These are all risk factors for musculoskeletal disorders. In addition, the surgical environment should be more ergonomic [4,7,8]. The incidence of musculoskeletal disorders has been studied in some subspecialities of orthopaedic surgery such as: traumatology, paediatric surgery, in residents, and among surgeons performing arthroplasty [2–4,9–12]. The prevalence of work-related musculoskeletal pain has reached rates as high as 96.5% in arthroplasty surgeons [2]. Despite the growing concern in recent decades, to the best of our knowledge, no studies have provided information on the ergonomic habits and risk postures of orthopaedic surgeons at work.

Total knee arthroplasty (TKA) is one of the most common surgical procedures in orthopaedics, and it must be performed ergonomically and safely. In Spain, more than

65,000 total knee arthroplasties are performed annually (138.6 per 100,000 inhabitants in 2020) [13]. When performing a homolateral TKA with the surgeon's dominant hand, there are basically two positions: facing or on the same side of the knee to be operated on. However, when performing a total knee replacement contralateral to the surgeon's dominant hand, there are different positions that the surgeon tries to accommodate. We have not found any studies that define or investigate the different positions the surgeon takes when performing this procedure.

Our study aims to determine which positions are ergonomically more favourable when orthopaedic surgeons perform contralateral knee arthroplasty using the dominant hand. Moreover, it aims to identify the surgical steps that require ergonomic corrections.

2. Materials and Methods

The first step in our research was to define four of the most common positions an orthopaedic surgeon can adopt to perform a TKA contralateral to their dominant hand. We identified the positions based on the most frequent and representative percentage of those used by surgeons in our environment. As shown in Figure 1, position A is on the opposite side of the knee to be operated on; position B is on the same side as the knee to be operated on; position C is with the patient's legs separated and the surgeon standing between them; and position D is facing the knee to be operated on, at the patient's feet.

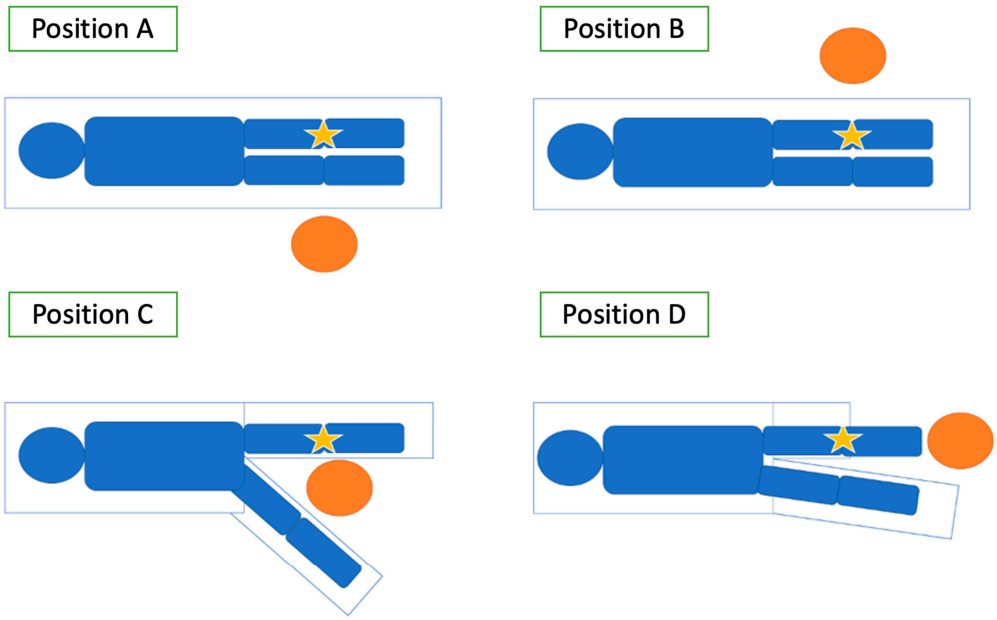


Figure 1. Positions for performing TKA surgery. This figure shows the different positions the surgeon can take to perform a knee replacement on the opposite side of the dominant hand. The circle represents the surgeon, and the star represents the knee to be operated on.

The next step was to define the main steps of TKA surgery to compare the different postures adopted in each position. To define these main steps, we took as a model the clinical practice of physicians in an orthopaedic surgery department of a hospital in the Levante region of Spain, where posterior stabilised TKAs are mainly implanted, and patella replacement is not performed. The critical steps defined were (1) incision (I), (2) patella eversion (PE), (3) anterior and posterior cruciate ligaments and meniscus release (CMR), (4) tibial-cutting guide positioning (TCG), (5) tibial cutting (TC), (6) femoral-cutting guide positioning (FCG), (7) distal femoral cut (DFC), (8) anteroposterior femoral cut (AFC), (9) lavage (L), (10) placement of the definitive femoral implant (DFI), (11) placement of the definitive tibial implant (DTI), and (12) closure (C).

2.1. Configuration and Instrumentation of the Recording Space

Our study is an in vitro assay with a human model. It involves human subjects: a patient model and a surgeon model, as well as observers, all physicians, and co-authors of the study. This type of experiment is exempt from the need for ethics committee approval in our country.

A right-handed surgeon, 180 cm tall and weighing 75 kg, with an arm length of 73 cm from the shoulder to the tip of the third finger, performed a simulation of a left-knee replacement in each defined position (A, B, C, and D) on a patient, 189 cm tall and weighing 95 kg, on an operating table with fixed tabletops (OPT/80, OPT SurgiSystems, Calliano, Italy), 200 cm long, in an operating room. For position C, the table allowed abduction of the unoperated leg. The working height was set at the beginning of the simulation (72 cm) as a comfortable working height for the surgeon's height, with the patient's knee in maximum flexion. The height of the table (as is usual in the operating theatre) was not modified during each simulation.

2.2. Images Capture for Angles Measurement

Next, we captured four video registrations to simulate left TKA surgery performed by a right-handed surgeon in the abovementioned positions. We obtained twelve images of each position corresponding to the twelve critical steps described. Subsequently, we measured the angles required for applying the physical workload assessment method Rapid Entire Body Assessment (REBA), using Kinovea software (v. 0.9.5). Kinovea is a completely free and open-source (software under the GPL v2 license) video-annotation tool, designed for sports analysis and developed by J. Charmant (asso@kinovea.org), that can be used to measure kinematic parameters. It is a valid and reliable tool that can measure accurately at distances up to 5 m from the object and in an angular range of 90° to 45° [14].

Four videos were recorded from the orthogonal perspective of the measurement angle to simulate a right-handed surgeon performing a left TKA in positions A, B, C, and D. Each position yielded twelve orthogonal images, or at least 90° and 60°, corresponding to the twelve critical phases described. Two experienced observers then measured the angles using Kinovea software.

2.3. Rapid Entire Body Assessment (REBA)

We used REBA (Rapid Entire Body Assessment) to assess the angles measured with Kinovea. The REBA method [15] is a tool for assessing postural workload. It arose in the search for a sensitive model to assess the forced postures that frequently occur in handling people (healthcare workers and carers). However, it is equally applicable to any sector or activity. Its main objective is to carry out a sensitive postural analysis for musculoskeletal risks; in other words, it informs us of the risk of a musculoskeletal disorder related to a particular posture. For this purpose, it divides the body into areas or segments that are coded individually. It considers correction factors for static postures, repetitive movements, unstable postures, and sudden postural changes. It also assesses the type of load grip. Finally, it results in a score that tells us whether preventive action needs to be taken and at which level of urgency.

For its application, this method divides the body segments into two groups: Group (A), where the neck, trunk, and legs are analysed, and Group (B), where the arms, forearms, and wrists are analysed (Figure 2). It provides a series of tables with schematic images that guide the researcher through obtaining a score for each segment and an overall score for Groups A and B. By applying REBA's conversion table, which relates the score of Group A to that of Group B, we obtain a final score. This score can be corrected for activity factors (static postures, repetitive movements, abrupt postural changes, or unstable postures). Thus, the final score will indicate the action level for each step of surgery, as shown in Table 1.

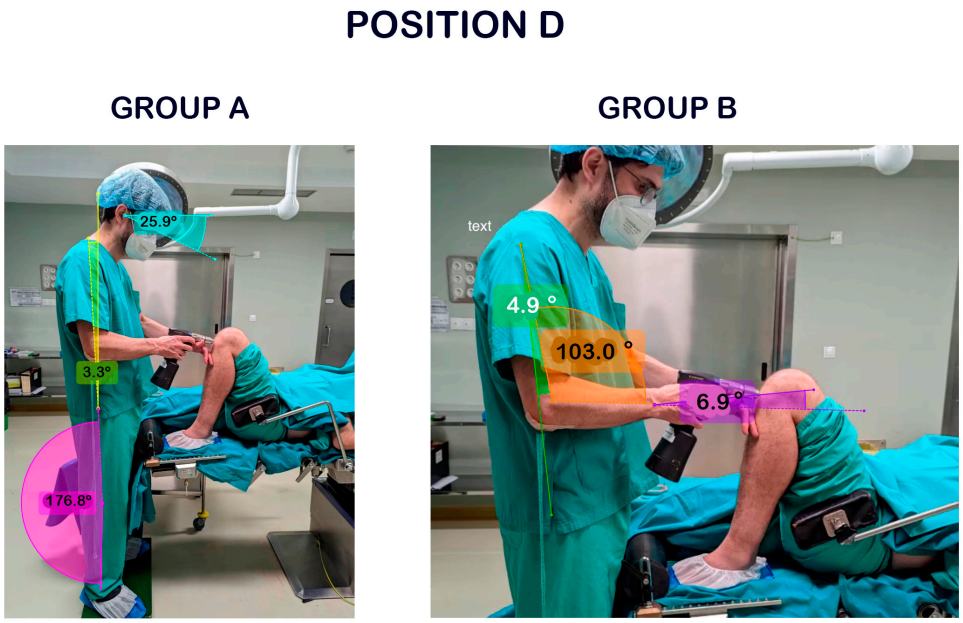


Figure 2. Rapid Entire Body Assessment (REBA) method. Example showing the application of the REBA method in the tibial-cut step in the D position with height modifications. We observe the measurements of Groups A and B.

Table 1. Action to take according to the final score of the Rapid Entire Body Assessment method.

Score	Risk Level	Level of Preventive Action
1	Negligible	Non-necessary
2–3	Low	May be necessary
4–7	Medium	Necessary
8–10	High	Necessary with priority
11–15	Very high	Necessary NOW

To evaluate each posture globally and to be able to compare them with each other, we added up the final scores of all the steps for each posture and divided them by 12 (the number of steps studied). Finally, by applying the REBA method, we obtained a final score for each of the 12 steps for each of the four defined positions. These scores allowed us to detect, for each position, which steps required early preventive action.

2.4. Kinovea and REBA Observers' Agreement

Angular measurements with Kinovea and REBA method application were performed twice for one observer over seven to ten days and once for a different trained observer. Before the measurements were taken, both evaluators agreed to the following:

(A) To define the measurement points for the angles (cervical tilt, trunk, legs, arm, forearm, and wrist);

(B) To take into account the following points for the REBA application:

- (1) To measure the most unfavourable posture in each step, with measurements always being taken on the working arm at each step;
- (2) To consider that the trunk is upright if it is <5° and that the wrist is deviated if it is >15°;
- (3) To consider that the neck and trunk are rotated if they are not directed to where the feet are looking;
- (4) To consider that the arm is in internal rotation if the hand is not placed in the same line as the elbow but more to the body's centre;

(5) To apply the static-posture correction factor in all steps;
(6) To apply the repetitive-action correction factor in steps where a saw or a hammer is used, i.e., bone cutting or implantation of prosthetic material is performed;
(7) To consider the neutral wrist angle since grasping the instruments does not require flexion or extension of the wrist. However, the need for lateral wrist deviation was considered a correction factor, depending on the activity and position of the surgeon.

2.5. Data Analysis

Statistical analysis was performed using the Statistical Package for the Social Sciences (SPSS), version 22 for Windows (SPSS, Inc., Chicago, IL, USA), establishing an α error of 5%, a confidence level of 95%, and a p-value of 0.05.

The angular values and REBA score were presented as mean and standard deviation (SD). For the analysis of this quantitative variable, we used Kolmogorov–Smirnov and Student's t-test or nonparametric alternatives for related samples. Finally, a simple comparison was made between Observer 1's first set of observations (1.1) and Observer 2's single round of measurements using the Wilcoxon test.

Test–retest reliability (reproducibility) and inter-rater analysis of Kinovea's angles and REBA were performed using the intraclass correlation coefficient of absolute concordance using a two-factor random-effects model [ICC (2,1)] [16]. We analysed the concordance of the torsion/rotation variable in the application of the REBA using Cohen's Kappa index. We assessed intra- and inter-observer reliability and Kappa index according to the criteria established by Landis and Koch (<0 indicates no agreement, 0.00 to 0.20 indicates slight agreement, 0.21 to 0.40 indicates fair agreement, 0.41 to 0.60 indicates moderate agreement, 0.61 to 0.80 indicates substantial agreement, and 0.81 to 1.0 indicate almost perfect or perfect agreement) [17].

Finally, we used the average of the values of the REBA application in each step to give an overall value for each position so that the position with the lowest average values for each observer would be the most ergonomic position.

3. Results

Table 2 shows the results of the angular measurements with Kinovea. We found statistically significant differences between the angular values for the B and D positions. However, according to Landis and Koch's criteria [18], the ICC (2,1) values in the test–retest reliability and inter-rater analysis (Table 3) of Kinovea's angles show an almost perfect correlation for the angular values in all positions.

Table 2. Results of angular measurements with Kinovea.

Kinovea Angles	Observer 1.1 *	Observer 1.2	Observer 2	Wilcoxon Obs. 1.1 vs. Obs. 2
	Mean (SD)	Mean (SD)	Mean (SD)	
Position A	61.93 (63.29)	62.56 (63.57)	61.83 (63.1)	0.559
Position B	59.85 (64.39)	59.91 (64.84)	56.28 (61.9)	0.016
Position C	60.81 (65.26)	61.28 (64.98)	59.8 (63.66)	0.12
Position D	59.14 (65.82)	56.6 (64.9)	54.71 (64.54)	0.002

* Observer 1.1: First assessment of the first observer. Observer 1.2: Second assessment of the first observer. Observer 2: Result of the second observer's assessment. SD: Standard deviation.

Table 3. Intraclass correlation coefficient of absolute concordance (ICC (2,1)) values for the test–retest reliability (Observer 1.1 vs. Observer 1.2) and for the inter-rater analysis (Observer 1.1 vs. Observer 2).

Kinovea Angles	ICC (2,1) Test–Retest Reliability (95% CI)	ICC (2,1) Inter-Rater Analysis (95% CI)
Position A	0.996 (0.994–0.998)	0.995 (0.991–0.997)
Position B	0.996 (0.994–0.997)	0.977 (0.963–0.986)
Position C	0.996 (0.994–0.997)	0.994 (0.991–0.997)
Position D	0.946 (0.915–0.966)	0.943 (0.909–0.964)

Table 4 shows the overall scores obtained by each observer using the REBA method. We found statistically significant differences between the REBA scores for the B position.

Table 4. Overall scores obtained by each observer using the REBA method and Kappa index for agreement of the torsion/rotation test–retest and interobserver.

REBA Scores		Observer 1.1 *	Observer 1.2	Observer 2	Wilcoxon Obs. 1.1 vs. Obs. 2
Position A	Mean (SD)	7 (2.33)	6.58 (1.97)	6.5 (2.23)	0.107
	Kappa	Test–retest: 0.804		Interobserver: 0.943	
Position B	Mean (SD)	6.17 (2.58)	6.17 (2.552)	5.5 (2.505)	0.033
	Kappa	Test–retest: 0.801		Interobserver: 0.885	
Position C	Mean (SD)	5.67 (2.015)	5.92 (2.151)	5.5 (1.931)	0.317
	Kappa	Test–retest: 0.776		Interobserver: 0.858	
Position D	Mean (SD)	3.58 (1.24)	3.75 (1.545)	3.42 (1.311)	0.48
	Kappa	Test–retest: 1		Interobserver: 0.62	

* Observer 1.1: First assessment of the first observer. Observer 1.2: Second assessment of the first observer. Observer 2: Result of the second observer's assessment. SD: Standard deviation.

The Kappa index, which analyses the agreement of the torsion/rotation variable when applying the REBA, showed almost perfect inter-rater agreement for positions A, B, and C, with values between 0.85 and 0.94, indicating substantial agreement for position D (0.62). The test–retest reliability showed substantial agreement for all items except for item D, where the value showed perfect agreement.

On the other hand, the ICC (2,1) values for the test–retest reliability and for the interobserver analysis (Table 5) of the REBA score show an almost perfect correlation for the angular values in all positions, according to Landis and Koch's criteria [18], except for the interobserver analysis of position D, for which the ICC (2,1) was 0.79, indicating substantial agreement.

Table 5. Intraclass correlation coefficient of absolute concordance (ICC (2,1)) values for the test–retest reliability and for the interobserver analysis.

REBA Score	ICC (2,1) Test–Retest Reliability (95% CI)	ICC (2,1) Inter-Rater Analysis (95% CI)
Position A	0.814 (0.492–0.942)	0.89 (0.657–0.967)
Position B	0.962 (0.873–0.989)	0.913 (0.614–0.977)
Position C	0.971 (0.892–0.992)	0.957 (0.865–0.987)
Position D	0.915 (0.743–0.974)	0.793 (0.436–0.935)

Figure 3 shows the results obtained after applying the REBA method. The position with the least ergonomic risk for the surgeon is option D, where the surgeon faces the

operated knee. The lowest postural risk would be D < C < B < A. Based on the average REBA score of all their measurements, the surgical steps with the highest ergonomic risk were the TCG, the FCG, the DFC, the AFC, and the DTI (Figure 4).

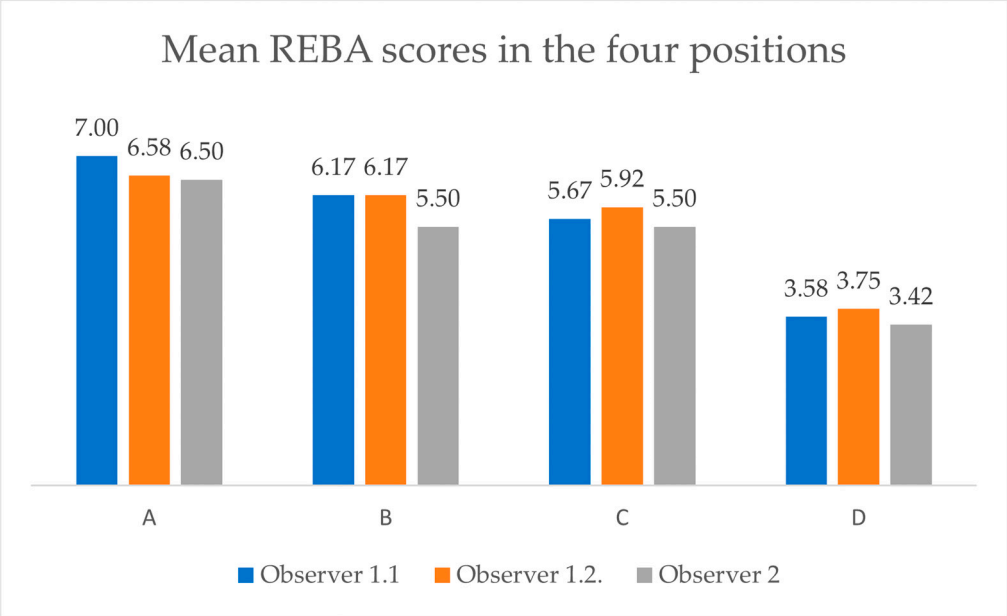


Figure 3. Results with the REBA method. Positions and average REBA score. A: Surgeon on the opposite side of the knee to be operated on. B: Surgeon on the same side as the knee to be operated on. C: Surgeon between both of the patient's legs. D: Surgeon facing the knee to be operated on, at the patient's feet. Observer 1.1: First assessment of the first observer. Observer 1.2: Second assessment of the first observer. Observer 2: Result of the second observer's assessment.

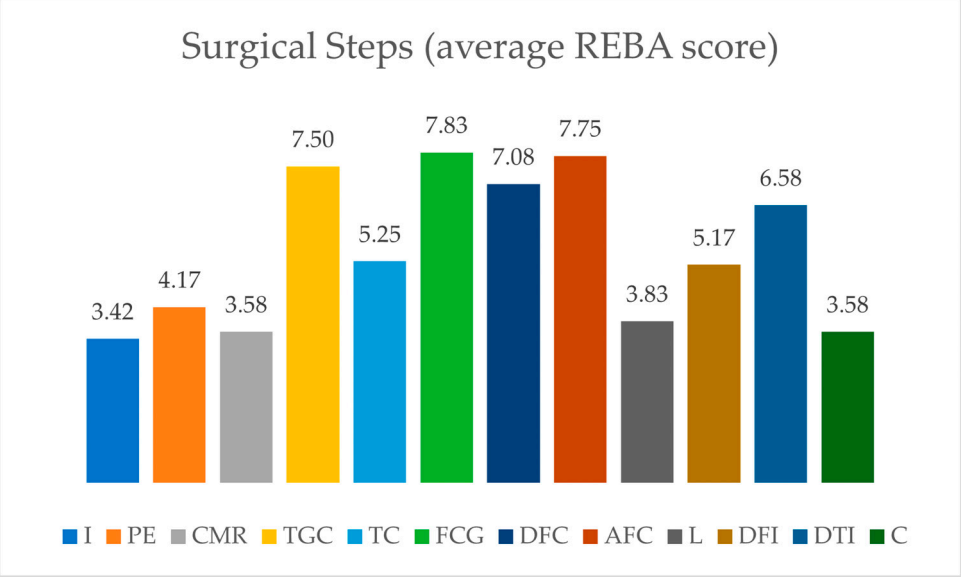


Figure 4. Results with the REBA method. Positions and average REBA score. Surgery steps: incision (I), patella eversion (PE), crusader and meniscus release (CMR), tibial-cutting guide (TCG), tibial cutting (TC), femoral-cutting guide (FCG), distal femoral cut (DFC), anteroposterior femoral cut (AFC), lavage (L), placement of the definitive femoral implant (DFI), placement of the definitive tibial implant (DTI), and closure (C).

4. Discussion

Our study aims to make the first approach to the problem of surgeon ergonomics during knee arthroplasty. To the best of our knowledge, our research is the first study to define and evaluate the different positions adopted by orthopaedic surgeons when performing TKA from an ergonomic perspective. Therefore, we cannot compare our results to other studies. As the main result of our study, the position with the least ergonomic risk for the surgeon is option D, where the surgeon faces the operated knee. The lowest postural risk would be D < C < B < A.

From the point of view of preventing these musculoskeletal disorders, we did not find any specific measures to reduce the risk for surgeons. However, some studies aim to study and extrapolate the ergonomic recommendations made in other work areas [6,18]. According to the Occupational Safety and Health Administration (OSHA), workers should be in a workspace that allows them to work at 15% or less of their maximum capacity. In the case of surgeons, achieving these goals requires them to maintain a neutral posture of their axis and limbs for most of the procedure. To achieve this, the surgeon must have an upright neck and trunk posture, avoiding flexion–extension and torsion; shoulders at 0° with no flexion–extension, abduction, or rotation; elbows as close to 90° as possible; and wrists avoiding flexion–extension and lateral deviation (radial or ulnar) [6].

In our study, the postures with the highest REBA scores did not meet the above requirements, resulting in a higher postural strain and ergonomic risk for the surgeons. This could explain why position A, where the surgeon is furthest from the work area, has the highest score after applying the REBA method.

In a study in which electromyographic measurements of specific muscle groups were taken for two surgeons to compare the differences in the effectiveness of operations performed on different sides, Ceyhan et al. [12] concluded that right-handed surgeons exert more effort when performing left-sided TKA and that left-sided knee operations cause more occupational injuries to surgeons.

One element we did not include in our study, but which plays a fundamental role in achieving these positions, is the height of the worktable or desk. The relationship between working height and ergonomic risk has been extensively studied in other occupations [19–21] and the surgical field [22,23], particularly in laparoscopic surgery [24,25].

Alaqeel et al. suggest different working heights depending on the type of activity the surgeon is performing [6]. A table height of approximately 5 cm below the elbow is recommended for tissue dissection. For more straightforward tasks, such as suturing and screw threading, the table should be approximately 5 to 10 cm below the elbow. However, for heavy tasks that require force on the floor (drilling, sawing, or hammering), it is recommended to work at a height of between 20 cm and 40 cm below the elbow. In our study, the surgeon set the table height to his comfort level (73 cm) at the start of the simulation, and no changes were made to the table height, as is customary during knee arthroplasty surgery, even though the working plane changed when the knee position was changed. Usually, with this table height, the surgeon mainly controls the knee in 90° flexion and in maximum flexion for tibial and femoral cuts (TCG, FCG, DFC, AFC) and component implantation (DFI, DTI), and it is usually less comfortable for steps with the knee in extension (I, PE, C), which tend to be the least performed. However, the steps with the highest ergonomic risk according to the average REBA were precisely those of cutting and implant placement. This implies that orthopaedic surgeons do not know and, therefore, do not consider the recommendations described by Alaqeel et al. regarding the height of the work surface in relation to the activity to be carried out. Intraoperative adaptation of the height of the table and of the surgeon by means of elevations according to the working plane and the activity to be performed could improve the REBA score and, thus, the ergonomic risk.

However, the surgeon's position during surgery is not the only element responsible for producing changes in the musculoskeletal system, and various authors have studied other factors. So, other recommendations that orthopaedic surgeons should know and

apply are to take microbreaks of 30–90 s every 20–40 min of work, as the literature suggests that these reduce the incidence of musculoskeletal disorders and increase concentration, improving performance without significantly increasing surgical time [26,27]. On the other hand, several studies propose specific physical training programmes for strength, endurance, and flexibility to reduce musculoskeletal disorders resulting from surgical work in different specialities [22,28,29]. Finally, most articles proposing measures to prevent work-related musculoskeletal disorders in surgeons agree on the need for ergonomic education programmes and increased awareness among professionals about the importance of such prevention [1,22,30], a need that is currently unmet [31].

As we can see from the literature, a very high percentage of orthopaedic surgeons have work-related musculoskeletal disorders [2,3,32]. Therefore, an ergonomic study of the different activities of the speciality, the development of preventive measures to avoid disability associated with these disorders, and an awareness of the importance of studying this field among surgeons are priorities.

There are some limitations to our study. Firstly, we use the average of the values from the REBA application in each step to give a global value for each position. This system has yet to be described or used in any other article in the bibliography. However, it makes sense and is reproducible. Secondly, only some of the material used in actual surgery was available during the simulations, as it would have been necessary to sterilise several boxes of surgical equipment, with the associated resource and time implications. However, we used the heaviest material in surgery: the motor and the hammer (Figure 5).



Figure 5. The heaviest surgical material used during total knee arthroplasty surgery. The weight of both tools is shown in milligrams.

Thirdly, we studied the ergonomics of different positions in the same surgeon on the same patient. Differences in the body morphology of each surgeon may influence each position's ergonomic risk. There is no strict extrapolation of our results to other morphometric parameters of the surgeon and the patient. Nevertheless, we used a "typical" surgeon as a model: a mesomorphic male, 1.80 m tall and weighing 75 kg. Fourthly, the recording of the simulations did not always provide orthogonal images for measuring the angles. However, each position yielded twelve orthogonal images, or at least 90° and 60°, corresponding to the twelve critical phases described, and Kinovea is a valid and reliable tool that can measure accurately at distances up to 5 m from the object and in an angular range of 90° to 45° [14]. Regarding the ability of the REBA method and our application of it to assess the ergonomic risk of each surgical posture, the results show that, despite the experience and previous agreement on the part of the evaluators to reduce the variability of the measurements, there may be minor variations in the angular measurements in some postures. There may be variations in the interpretation of the torsion/rotation, which is also involved in the result of the REBA application. However, after assessing the intra- and inter-observer variability, we found that the REBA application is a reproducible and valid procedure. Fourthly, our analysis was performed assuming a "tibia-first technique" with mechanical alignment and conventional instrumentation, so our results cannot be extrapolated to techniques with initial femoral osteotomies, techniques with nonmechanical alignment, computer-, PSI-, or robotic-assisted surgery. In addition, factors such as the

use of space suits during surgery, the height of the operating table, the position of the scrub nurse, or the possibility of the assistant surgeon performing the most uncomfortable steps for the leading surgeon were not considered. The control and measurement of these variables and their influence on surgeon ergonomics could be the focus of future studies.

5. Conclusions

In light of our study results, we recommend option D, in which the right-handed surgeon faces the left knee to perform TKA surgery, as the position with the lowest ergonomic risk of musculoskeletal injuries in left TKA surgery. Nevertheless, the REBA score indicates that actions are needed to reduce the risk of musculoskeletal disorders further.

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